

Series Program:-

1) Print Series (1-2 3-4 5-6 n)

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int n;
```

```
    cout << "Enter Number";
```

```
    cin >> n;
```

```
    for (int i=1; i<=n; i++)
```

```
        if (i%2==0)
```

```
            cout << -i << " ";
```

```
        else
```

```
            cout << i << " ";
```

```
    }
```

2 Series with Power and Factorial combined

$$\left(\frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right)$$

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int n;
```

```
    float x, sum=1, term=1;
```


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```
cout << "Enter value of x";  
cin >> x;  
cout << "Enter number of Series";  
cin >> n;  
for (int i = 1; i <= n; i++)  
    term = term * x / i;  
    sum = sum + term;  
cout << "Sum of Series" << sum;  
}
```

3 Sine Series (Taylor Series)

$$\sin(n) = n - \frac{n^3}{3!} + \frac{n^5}{5!} - \dots$$

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int n;
```

```
    float sum = 0; term;
```

```
    cout << "Enter value of x (in radian): ";
```

```
    cin >> x;
```

```
    cout << "Enter Number of term: ";
```

```
    cin >> n;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        int power = 2 * i - 1; // 1, 3, 5, ...
```


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// calculate Power of x

float num = 1;

for (int j = 1; j <= power; j++)

num = num * x;

// calculate factorial

int fact = 1;

for (int j = 1; j <= power; j++)

fact = fact * j;

term = num / fact;

// Alternating sign

if (i % 2 == 0)

sum = sum - term;

else

sum = sum + term;

}

cout << "sum(x) = " << sum;

}

4) **Triangular Series:** 1 3 6 10 15

#include <iostream>

using namespace std;

int main()

{

int n;


```
cout << "Enter Number";
```

```
cin >> n;
```

```
for (int i=1; i<=n; i++)
```

```
    j = j * (i+1) / 2;    // j = j + 1;    j = 0;
```

```
    cout << j << " ";
```

```
}
```

6 Geometric Series:-

1 + 2 + 4 + 8 + 16 +

```
#include <iostream>
```

```
using namespace std;
```

```
int main ()
```

```
{
```

```
    int n;    float n1, term, sum = 0;
```

```
    cout << "Enter 1 Number";
```

```
    cin >> n1;
```

```
    for (int i=1; i<=n; i++)
```

```
        cout << "Enter common Ratio";
```

```
        cin >> r;
```

```
        cout << "Enter Number of terms:";
```

```
        cin >> n;
```

```
        term = n1;
```

```
        for (int i=1; i<=n; i++)
```

```
{
```



```

    sum = sum + term;
    cout << "term = " << term << " ";
    term = term * n;
}
cout << "\n Sum = " << sum;
}

```

7 Harmonic Series :-

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n}$$

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int n;
```

```
    double sum = 0;
```

```
    cout << "Enter Number of term";
```

```
    cin >> n;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        sum = sum + 1.0/i;
```

```
        cout << "1/" << i << " ";
```

```
    }
```

```
    cout << "\n Sum = " << sum;
```

```
}
```


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8 Mixed Series :-

$$2^2 + 3^3 + 4^2 + 5^3 + \dots$$

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int n, sum=0;
```

```
    cout << "Enter number of term";
```

```
    cin >> n;
```

```
    for (int i=2; i<=n; i++)
```

```
    {
```

```
        if (i%2==0)
```

```
            sum = sum + (i*i);
```

```
            cout << "i*i << " << " ";
```

```
        else
```

```
            sum = sum + (i*i*i);
```

```
            cout << "i*i*i << " << " ";
```

```
    }
```

```
    cout << "\n Sum" << sum;
```

```
}
```


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9) Pattern Series :-

1
1 2
1 2 3
1 2 3 4

```
#include <iostream>
```

```
using namespace std;
```

```
int main ()
```

```
{
```

```
    int n;
```

```
    cout << "Enter number of rows";
```

```
    cin >> n;
```

```
    for (int i = 1; i <= n; i++)
```

```
    {
```

```
        for (int j = 1; j <= i; j++)
```

```
            cout << j;
```

```
        }
```

```
        cout << endl;
```

```
    }
```

10) Prime Number :-

2, 3, 5, 7, 11, 13, 17, ...

```
#include <iostream>
```

```
using namespace std;
```

```
int main ()
```

```
{
```

```
    int n, count = 0, num = 2;
```


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```
count << "Enter number of term";  
cin >> n;  
while (count < n)  
    int isPrime = 1;  
    for (int i = 2; i <= num / 2; i++) {  
        if (num % i == 0)  
            isPrime = 0;  
        break;  
    }  
    if (isPrime == 1) {  
        cout << "num" << " ";  
        count++;  
    }  
    num++;  
}
```